

Level 0	Level 1	Level 2	Level 3	Level 4	Level 5
Driver support features			Automated driving features		
Warnings and momentary assistance	Steering or brake/acceleration support	Steering and brake/acceleration support	Self-driving activated under certain conditions		Self-driving at all times
- Automatic emergency braking - Blind spot warning	- Lane centering - OR - Adaptive cruise control	- Lane centering - AND - Adaptive cruise control	Traffic jam chauffeur	Local driverless taxi	Same as Level 4, but self-driving everywhere under any conditions
You are driving and must constantly supervise the driver support features			You are not driving when automated driving features are engaged		

SAE J3016 levels of driving automation (Source: Society of Automotive Engineers, 2019)

initially so that training data can be built into the autopilot application. Later it can become fully automated based on trained model and artificial intelligence system built into the application.

Tesla's autopilot AI

"The advantage that Tesla will have is that we'll have millions of cars in the field with full autonomy capability and no one else will have that." — Elon Musk

Tesla is considered as tech leader in the world of electric vehicles. With the integration of artificial intelligence techniques and smart sensors, various features like autopilot are integrated in cars nowadays. Tesla is making use of data science, neural networks, and deep learning for understanding the datasets of customers to gather and predict information about customers and improve the vehicles' functionalities.

Tesla vehicles have advanced AI features that not only look futuristic but magical in terms of technology. Some of these are described below briefly.

AI integrated chips. Tesla creates AI integrated chips for assistance in lane mapping, autonomous navigation. Six billion transistors are integrated on a single Tesla chip and Tesla chips are 21 times faster than Nvidia chips having 32MB SRAM, which is faster as compared to DRAM.

Autopilot. Tesla integrated traffic-aware cruise control and autosteer features 360-degree visualisation. Tesla uses eight cameras with neural

networks to have 360-degree visualisation for safe driving.

Batteries and propulsion system.

Tesla uses lithium-ion batteries to pump in more energy and better transmission speeds.

Tesla's following features make autopilot possible:

- Radar - front facing
- High megapixel cameras - visibility upto 250m
- Digital controlled electric assistance braking system
- Twelve long-range ultrasonic sensors

Cruise control. The radar and forward-facing cameras track the position of cars ahead and adjust the Tesla's speed accordingly. This feature maintains a safe distance between you and the car in front. The distance between the cars depends on the speed at which both the cars are moving. If a car merges into your lane, the Tesla will monitor its position and reduce speed, if necessary.

Autosteer. The autosteer function lets the Tesla stay centered in a lane, change lanes, and self-park. To keep the Tesla centered, the cameras around the car track the positioning of road markings and the sensors monitor other cars on the road to keep a safe distance. To self-park in parallel and perpendicular positions the car uses both sensors and cameras to avoid hitting any surroundings. The autopilot feature does not work well when the road markings are not clear and/or when the car is traveling under 32kmph. This feature is not

recommended for use in residential zones where there are streetlights and stop signs.

Six levels of self-driving autonomy in autopilot applications

The concept of autonomy levels was derived by International Society of Automotive Engineers (SAE) in 2014 under 'Taxonomy and Definitions for Terms Related to On-Road Motor Vehicle Automated Driving Systems' report. The report highlighted six levels of self-driving autonomy that auto manufacturers need to achieve to attain no-steering wheel technology in the future. Later, in September 2016, the National Highway Traffic Safety Administration (NHTSA) adopted SAE standard.

The six levels of self-driving autonomy are:

Level 0: No Automation. It means the human driver is solely responsible for 100% driving task. It is not very hard to understand, and even till today there are modern cars with driver-assistance features and classified as Level 0. The car system can give some warnings like lane departure, forward collision, etc., and even inform the driver via notifications and beep warnings. These vehicles fall under Level 0 category.

Level 1: Driver Assistance. Under Level 1, in certain driving modes, the car can take over the control of steering wheel or pedals, but not both of them at the same time. Example of Level 1 can be adaptive cruise control, where the car will keep a set speed and distance safety with other vehicles on the road and automatically brake if the traffic slows down, and resumes the original speed once normal traffic continues. Another feature is lane assistance, which automatically brings the car to middle position in lane, if vehicle is observed off lane.

Level 2: Partial Automation. In this level, the car can steer, accelerate, and brake in certain conditions. Under this level, the driving task is shared between the vehicle and driver. The vehicle performs two primary driving functions - lateral and longitudinal control by combining adaptive cruise control and lane